$$\cos\alpha = \frac{2}{3},\ \cos\beta = \frac{2}{3},\ \cos\gamma = -\frac{1}{3},$$

where α, β, γ are the angles made by $\hat{b}$ with the positive x-axis, positive y-axis, and positive z-axis, respectively. As the angle between the given straight line with the coordinate axes are same as the angles made by $\hat{b}$ with the coordinate axes, we have $\alpha = \cos^{-1}\left(\frac{2}{3}\right)$, $\beta = \cos^{-1}\left(\frac{2}{3}\right)$, $\gamma = \cos^{-1}\left(\frac{-1}{3}\right)$, respectively.

6.7.4 Angle between two straight lines

(a) Vector form

The acute angle between two given straight lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ is same as that of the angle between $\vec{b}$ and $\vec{d}$. So, $\cos\theta = \frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}$ or $\theta = \cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right)$.

Remark

(i) The two given lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ are parallel

$\Leftrightarrow \theta = 0 \Leftrightarrow \cos\theta = 1 \Leftrightarrow |\vec{b}\cdot\vec{d}| = |\vec{b}|\,|\vec{d}|$.

(ii) The two given lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ are parallel if, and only if $\vec{b} = \lambda\vec{d}$, for some scalar λ.

(iii) The two given lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ are perpendicular if, and only if $\vec{b}\cdot\vec{d} = 0$.

(b) Cartesian form

If two lines are given in Cartesian form as $\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$ and $\frac{x-x_2}{d_1} = \frac{y-y_2}{d_2} = \frac{z-z_2}{d_3}$, then the acute angle θ between the two given lines is given by

$$\theta = \cos^{-1}\left(\frac{|b_1d_1 + b_2d_2 + b_3d_3|}{\sqrt{b_1^2 + b_2^2 + b_3^2}\sqrt{d_1^2 + d_2^2 + d_3^2}}\right)$$

Remark

(i) The two given lines with direction ratios b_1, b_2, b_3 and d_1, d_2, d_3 are parallel if, and only if $\frac{b_1}{d_1} = \frac{b_2}{d_2} = \frac{b_3}{d_3}$.

(ii) The two given lines with direction ratios b_1, b_2, b_3 and d_1, d_2, d_3 are perpendicular if and only if $b_1d_1 + b_2d_2 + b_3d_3 = 0$.

(iii) If the direction cosines of two given straight lines are l_1, m_1, n_1 and l_2, m_2, n_2, then the angle between the two given straight lines is $\cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|$.

12th_Maths_EM_Vol1_CH 6 Vector Algebra.indd 247 16-12-2022 14:25:19